

# Data Model

|                  |                            |
|------------------|----------------------------|
| Practical No.:   | PL-300 Practical 3         |
| Practical Title: | Design a Data Model        |
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## Project Summary

Went on to do the data model. It involves creating relationships between tables and then configuring table and column properties to improve the friendliness and usability of the data model. I also created hierarchies and create quick measures. Formatting was also done for numerical values to make the data to be more presentable.

## Objectives

List up to four objectives for this practical, each in 10 words or fewer.

| Relationships                                                                                                                                                    | Tables & columns                      | Hierarchies        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------|
| Create model relationships<br>The file was configured to not identify relationships between tables, which isn't the default setting hence they should be created | Configure table and column properties | Create hierarchies |

## Overview

List the steps you performed, in order. Adding a screenshot for each step makes the process easier to follow.

1. Created relationships

| From: table (column) ↑            | Relationship | To: table (column)                     | Status       |
|-----------------------------------|--------------|----------------------------------------|--------------|
| Σ Sales (EmployeeKey)             |              | Salesperson (Performance) (E...        | Inactive ... |
| Sales (ProductKey)                |              | Product (ProductKey)                   | Active ...   |
| Sales (ResellerKey)               |              | Reseller (ResellerKey)                 | Active ...   |
| Sales (SalesTerritoryKey)         |              | Region (SalesTerritoryKey)             | Active ...   |
| SalespersonRegion (Employee...    |              | Salesperson (Performance) (E...        | Active ...   |
| SalespersonRegion (SalesTerrit... |              | Salesperson (Performance) (EmployeeID) | Active ...   |
| Targets (EmployeeID)              |              | Salesperson (Performance) (E...        | Active ...   |

Figure 1: Final overview of the relationships

2. Configured the tables by:
- a. Adding a hierarchy
  - b. Adding a group for formatting

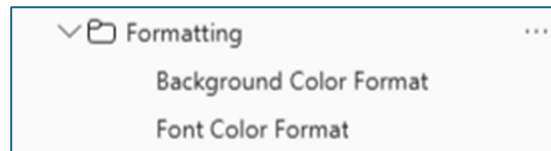


Figure 2: Formatting group was made

- c. Aligning to proper Data categories
- d. Formatting the columns was done

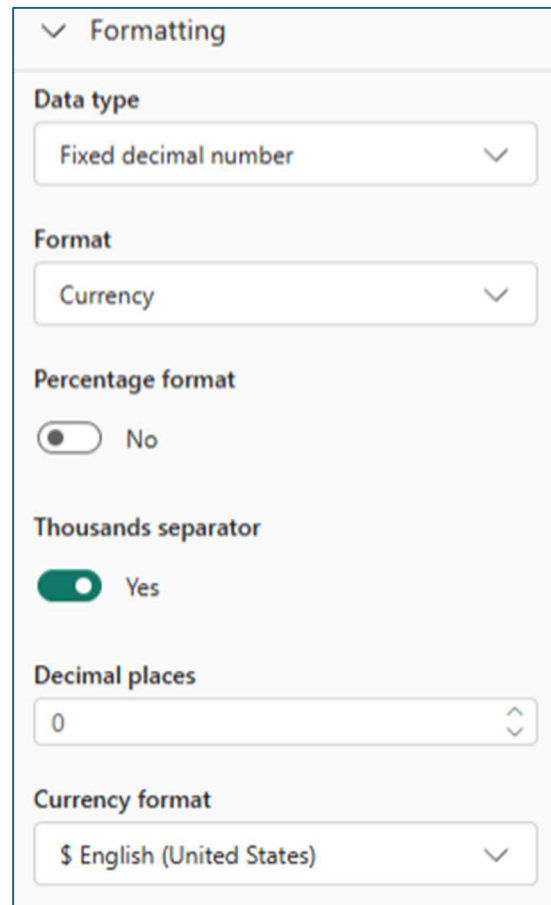


Figure 3: Example of formatting that was done on the Product | Standard cost

- 3. Columns with import data such as IDs were hidden (Not all were included in the table below):

| Table             | Column            |
|-------------------|-------------------|
| Region            | SalesTerritoryKey |
| Reseller          | ResellerKey       |
| Sales             | EmployeeKey       |
| Salesperson       | EmployeeID        |
| SalespersonRegion | EmployeeKey       |
| Targets           | EmployeeID        |

- 4. Disenabled Time Intelligence
- 5. Added quick measures:
  - a. Profit [Sales – Cost]
  - b. Profit Margins [Profit / Sales]
- 6. Worked around the one-to-many relationships and added the Targets table to model view

## Figures

| Category     | Sum of Sales         |
|--------------|----------------------|
| Accessories  | 539 106,09           |
| Bikes        | 64 069 033,16        |
| Clothing     | 1 714 056,05         |
| Components   | 11 226 374,90        |
| <b>Total</b> | <b>77 548 570,20</b> |

Figure 4a: Sales vs Category before formatting

| Category     | Sum of Sales       |
|--------------|--------------------|
| Accessories  | R539 106           |
| Bikes        | R64 069 033        |
| Clothing     | R1 714 056         |
| Components   | R11 226 375        |
| <b>Total</b> | <b>R77 548 570</b> |

Figure 4b: Sales vs Category before formatting

| Salesperson              | Sum of Sales       | Sum of TargetAmount  |
|--------------------------|--------------------|----------------------|
| Amy Alberts              | R10 288 626        | \$19 450 000         |
| Brian Welcker            | R77 548 570        | \$221 700 000        |
| David Campbell           | R12 004 822        | \$19 625 000         |
| Garrett Vargas           | R13 875 633        | \$23 675 000         |
| Jae Pak                  | R8 410 883         | \$13 575 000         |
| Jillian Carson           | R7 633 387         | \$13 675 000         |
| José Saraiva             | R13 875 633        | \$18 875 000         |
| Linda Mitchell           | R25 634 503        | \$40 850 000         |
| Lynn Tsoflias            | R1 391 025         | \$3 210 000          |
| Michael Blythe           | R21 987 348        | \$31 150 000         |
| Pamela Ansman-Wolfe      | R30 005 939        | \$53 850 000         |
| Rachel Valdez            | R1 877 743         | \$4 125 000          |
| Ranjit Varkey Chudukatil | R4 527 840         | \$9 050 000          |
| Shu Ito                  | R18 001 116        | \$59 850 000         |
| Stephen Jiang            | R65 868 919        | \$110 150 000        |
| Syed Abbas               | R1 391 025         | \$3 050 000          |
| Tete Mensa-Annan         | R12 004 822        | \$17 100 000         |
| Tsvi Reiter              | R7 638 607         | \$13 250 000         |
| <b>Total</b>             | <b>R77 548 570</b> | <b>\$676 210 000</b> |

Figure 5: Final sales person table summary

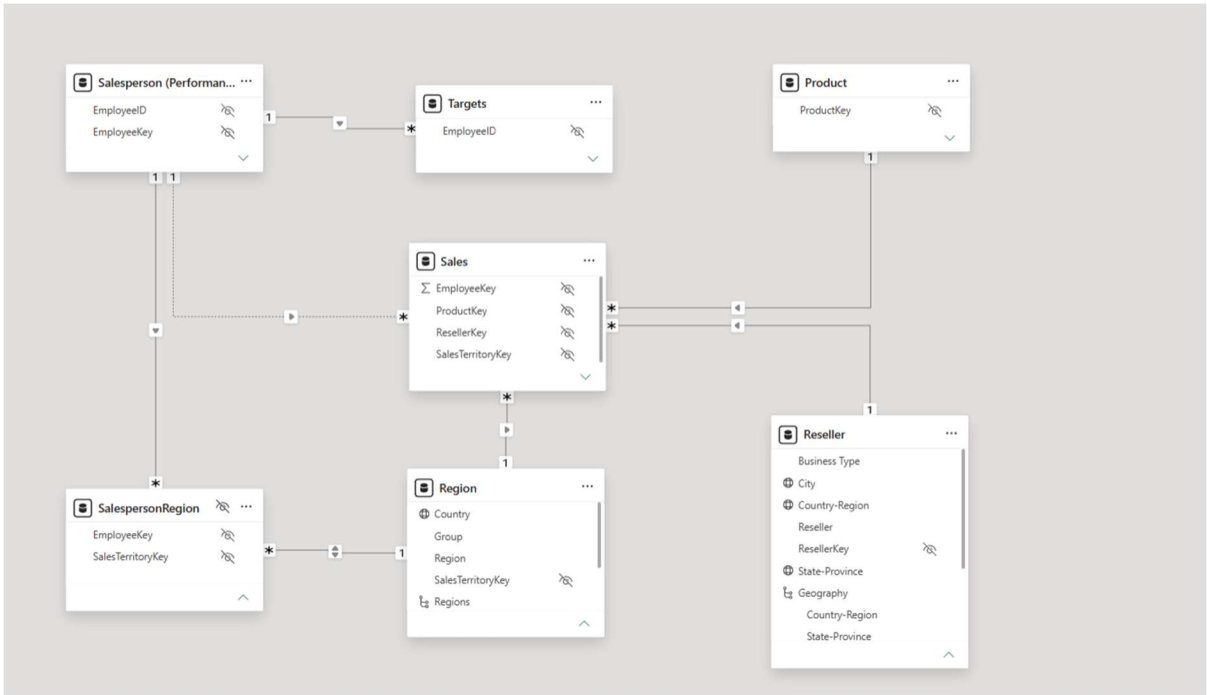


Figure 4: Final model view showing the relationship of the data